

Laboratory Activity 4

Assembly Language Programming Using EMU8086

OBJECTIVES:

At the end of the exercise, the students should be able to:

- familiarize with the different commands and tools used in EMU8086; and
- create assembly programs.

SOFTWARE REQUIREMENT:

- EMU8086 4.08

BASIC PRINCIPLES:

EMU8086 is a powerful tool for learning assembly language. It combines a source editor, assembler, disassembler, and emulator. It compiles the source code and executes it on emulator step-by-step. Its interface is very easy to work with. It allows you to watch registers, flags, and memory while your program is running. The emulator runs programs on a Virtual PC; this guarantees that your programs will not access real hardware. This also makes debugging much easier.

PROCEDURE:

1. Double-click the Desktop icon () to open/launch the emu8086, and then, click **New**. You will be asked to choose a code template as shown below.

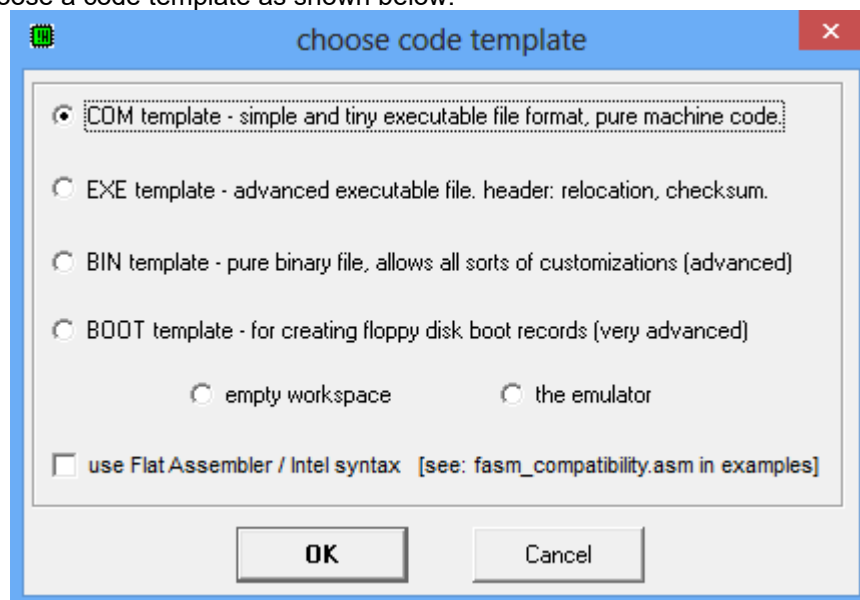


Figure 6.1

2. Click **OK**. The IDE will be displayed as shown in **Figure 6.2**.

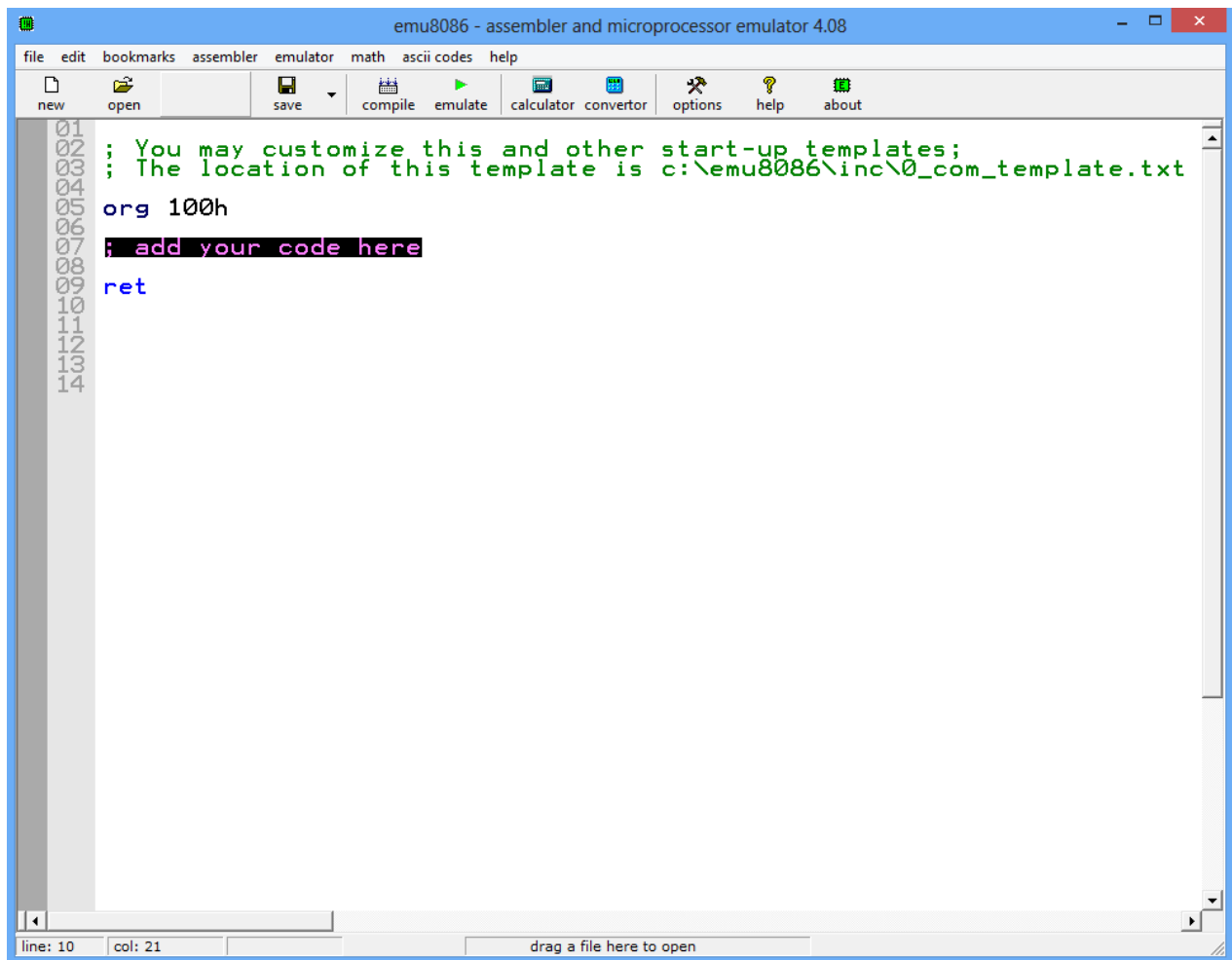


Figure 6.2

```

; You may customize this and other start-up templates;
; The location of this template is c:\emu8086\inc\0_com_template.txt
org 100h
; add your code here
ret

```

The statements after the semicolon (;) and in green are comments are ignored by the compiler. The line **org 100h** indicates that the program will begin at offset 100h. The ORG (which stands for origin) is an assembly directive that defines where the machine code is placed in memory. A directive is a command that has no influence on the operation of the microcontroller. The offset is always 100h for COM format. The command **ret** exits the program and returns you to the operating system.

3. Delete the default comments and write your own. Type a comment that this program will display the title of your favorite song.
4. Type **jmp start** below **org 100h**. This indicates an unconditional jump; this ensures the space for data so that it does not interfere with code and vice versa.
5. Declare a string by typing **msg: db "Song Title", 0Ah, 0Dh, 24h** below your previous line. Then, press Enter.

Explanation:

msg refers to your variable name

db is used for writing specified value into program memory

"Song Title" is the title of your song (characters that are enclosed in double quotation marks represent a string)

0Ah is equivalent to line feed in ASCII, which moves the cursor to the next row of the screen but stays in the same column

0Dh is equivalent to carriage return in ASCII which moves the cursor to the beginning of the current row

24h is the address for system interrupt controller (this is used to terminate the string)

6. Type **start: mov dx, msg** then press Enter. This line indicates that the offset of the variable *msg* will be stored in dx. The *start* in this line is the one mentioned in Step 4.
7. Bring your cursor just below the *mov* command. This is to better group the instructions.
8. Display the string by typing **mov ah, 09h** then press Enter. This is the command for print function.
9. Type **int 21h** then press Enter. This command ends the current process.
10. Type **mov ah, 0** then press Enter. This command is used to read an input from the keyboard.
11. Type **int 16h** then press Enter. This command waits for an input from the user in order to end the program. This permits the **ret** command to exit the program and return you to the operating system.
12. Click the **emulate** icon from the toolbar as shown below or simply press **F5**. The emulator window will appear (See Figure 4.4).

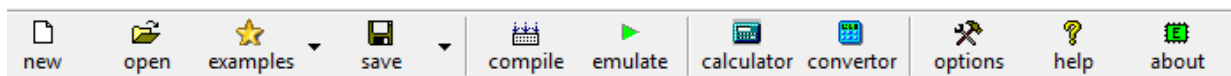


Figure 6.3

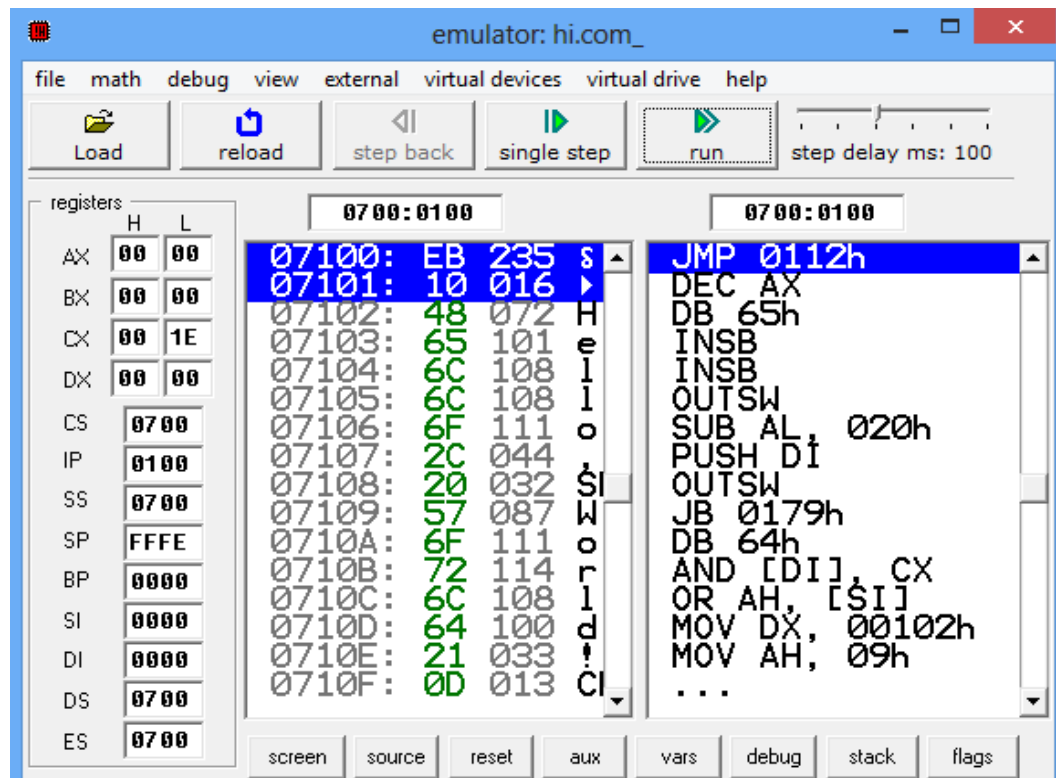
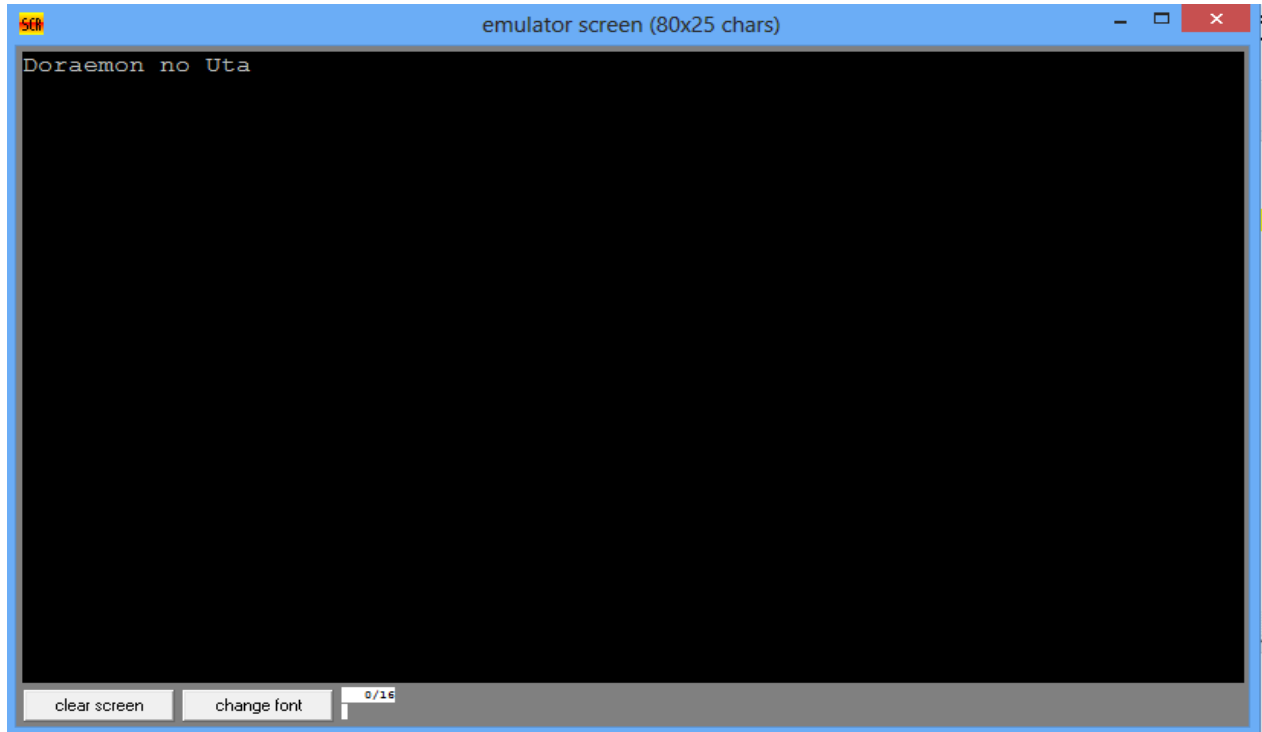


Figure 6.4

13. Click the **run** icon from the toolbar shown in Figure 46.4.

14. Press Enter to terminate the program.

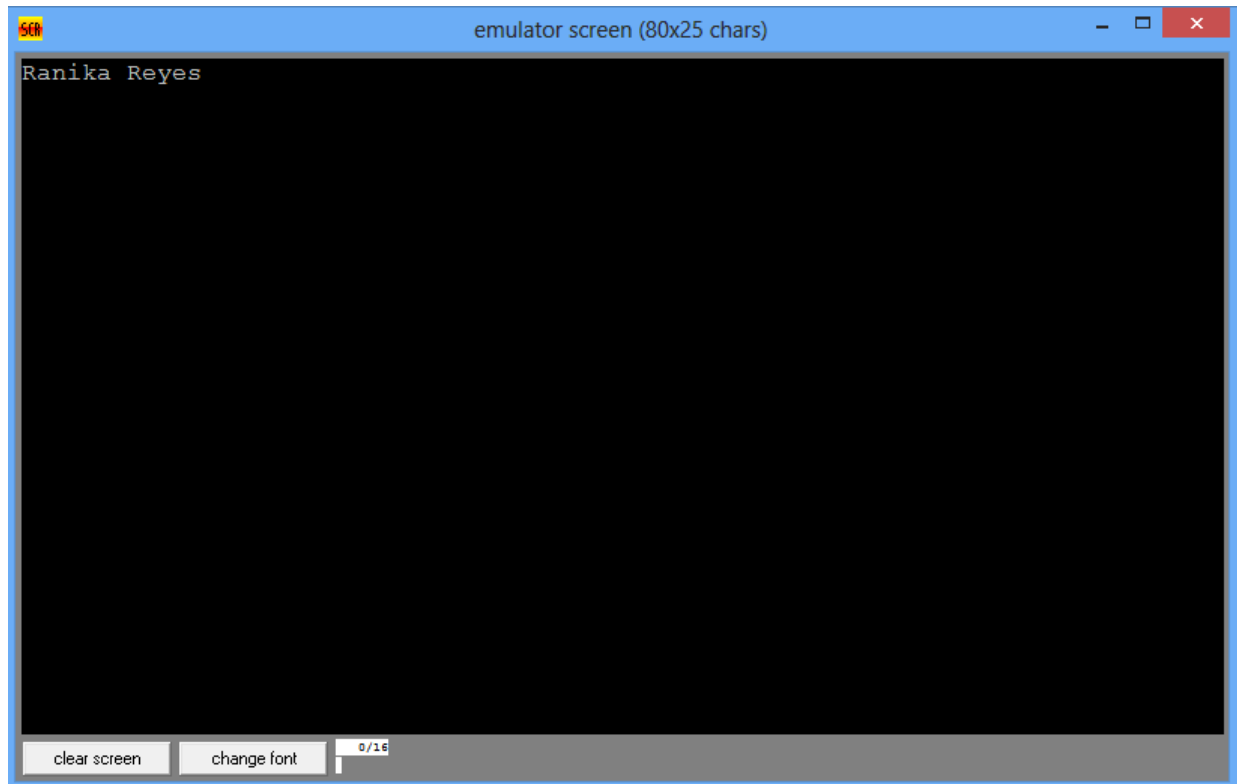
SAMPLE OUTPUT:



LABORATORY WORK

1. Open Camera and share your screen while doing the laboratory.
2. Create a new COM Template.
3. Display your first name and last name using two (2) string variables, one (1) to store your first name and one (1) to store your last name.
4. Save the program as **LabAct4.asm**.
5. Send your video link and screenshots of your output in the submission bin.

See Sample Output on the next page.



Photo/s of the Output:

